



Course Name: Microprocessor Programming & Interfacing
 Program: BS(EE)
 Duration: 60 Minutes
 Paper Date: 15-11-2018
 Section: ALL
 Exam Type: Mid Term - II

Course Code: EE303
 Semester: Fall 2018
 Total Marks: 30
 Weightage: 15%
 Page(s): 1

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Roll No. 15-4521

Section: EE-2

Instruction/Notes: Show complete working for each question, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Attempt each question only once.

Question#1:

A column of two push buttons is connected with 8051 microcontroller as shown in figure 1. Write an assembly language program that scans the push buttons (PB) and generates 1 kHz square wave at port pin P3.0 with varying duty cycle depending upon which push button is pressed. Table 1 shows the detail.

Assume that the oscillator frequency is 12MHz and the push buttons introduce bouncing for 30msec.

Status of Push Buttons	Duty Cycle
No button is pressed	100%
PB1 is pressed	50%
PB2 is pressed	75%

Table 1

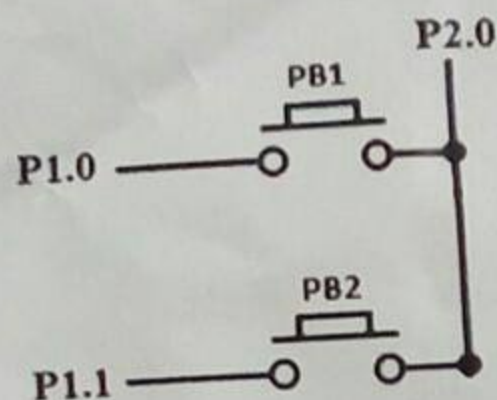


Figure 1

[15 Marks] CLO-03

Question#2:

Write an assembly language program that calculates factorial of a number 'n', varying from 0 to 5, a 3-bit number given as input via port P1, and displays the factorial result on 4-digit multiplexed common cathode 7-segment display. Formula to calculate factorial is given below:

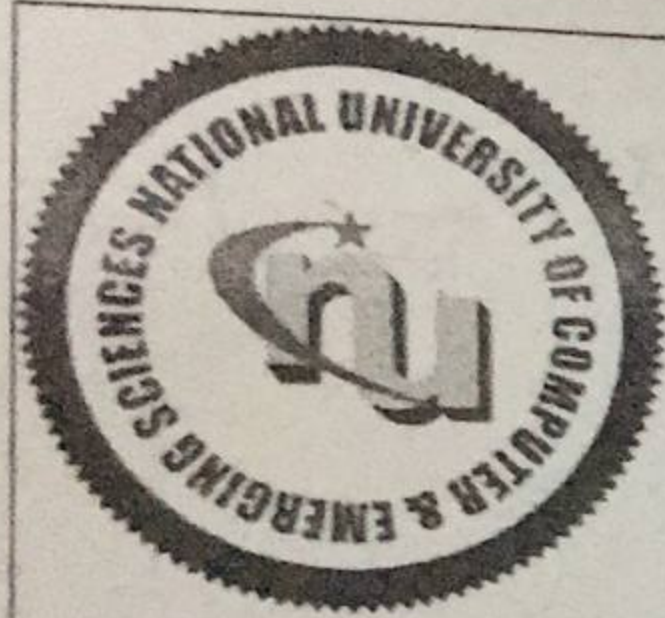
$$n! = n(n-1)(n-2) \dots (1)$$

Where,

$$0! = 1! = 1$$

The user will prepare input number over binary switches and then gives a positive pulse (0-1-0) on control switch connected to P1.4. Once the user give input i.e., value of 'n', the program will display the factorial result on 4-digit multiplexed common cathode 7-segment display. The switches introduce bouncing for 25msec. Your code must be as generic as possible. Assume that the delay routine for 25msec is already available with name "Delay_25ms".

National University of Computer and Emerging Sciences, Lahore Campus



Course:	MICROPROCESSOR INTERFACING & PROGRAMMING	Course Code:	EL-303
Program:	BS(Electrical Engineering)	Semester:	Fall 2017
Duration:	30 minutes	Total Marks:	20
Paper Date:	26-September-17	Weight	
Section:	EL 2A	Page(s):	1
Exam:	Quiz 1	Roll No:	L15-4522

Q.1 Write an assembly language program on KEIL and debug it. An LED is connected to one of the port pins of the 8051 microcontroller. The program is used to flash the LED continuously as in the following sequence:

3 flashes with ⁵⁰⁰~~250~~ms interval between each flash. ^{3s}

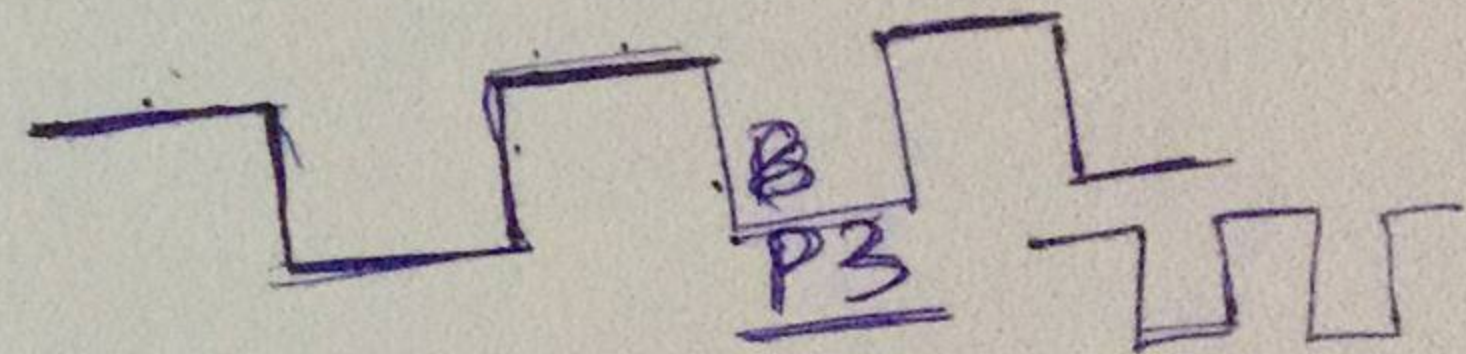
2 s delay.

3 flashes with ⁵⁰⁰~~250~~ms interval between each flash. ^{3s}

2 s delay..... This process is repeated forever.

$$f = 11.0592 \text{ MHz}$$

$$T = 1/f$$



$$500 \times 3$$

$$\boxed{2.5s} \approx 3s$$

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Course:	MICROPROCESSOR INTERFACING & PROGRAMMING	Course Code:	EL-303
Program:	BS(Electrical Engineering)	Semester:	Fall 2017
Duration:	30 minutes	Total Marks:	20
Paper Date:	25-Oct-17	Weight	
Section:	EL 3A	Page(s):	1
Exam:	Quiz 2	Roll No:	

Q.1 Write a program that takes two 4-bit numbers on Port 3 as input (Lower nibble is A, upper nibble is B). It multiplies the two digits and displays the answer in decimal base form on LEDs as shown in the following examples:

Example 1

If A = 5 and B = 8 then multiplication would give 225

display 5 on LEDs

1 sec delay

display 2 on LEDs

1 sec delay

display 2 on LEDs

1 sec delay

Example 2

If A = 1 and B = 2 then multiplication would give 2

display 2 on LEDs

1 sec delay

display 0 on LEDs

1 sec delay

display 0 on LEDs

1 sec delay

$$11.0592 \text{ MHz} \times 10^{-6}$$

$$MC = 1.085 \times 10^6$$

$$\frac{15}{1.085 \times 10^{-6}} = \frac{922 \times 10^3}{255} \text{ 1st}$$

$$\frac{3615}{255} = 15$$

$$\begin{array}{r} 22 \\ 10 \overline{) 225} \\ \underline{20} \\ 25 \end{array}$$

$$\begin{array}{r} 25 \\ 10 \overline{) 25} \\ \underline{20} \\ 5 \end{array}$$

$$\begin{array}{r} 225 \\ 10 \overline{) 225} \\ \underline{20} \\ 25 \\ \underline{20} \\ 5 \\ \underline{5} \\ 0 \end{array}$$

$$\begin{array}{r} 15 \\ 10 \overline{) 156} \\ \underline{10} \\ 56 \\ \underline{50} \\ 6 \end{array}$$

$$08H \rightarrow 5$$



Course Name:	Microprocessor Programming & Interfacing	Course Code:	EE303
Program:	BS(EE)	Semester:	Fall 2017
Duration:	60 Minutes	Total Marks:	30
Paper Date:	03-11-2017	Weightage	15%
Section:	ALL	Page(s):	2
Exam Type:	Mid Term - 2		

Student : Name: Ayesha Javed

Roll No. L15-4522

Section: EE-2

Instruction/Notes: Show complete working for each question, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Attempt each question only once.

Question#1: [15 Marks] CLO-03

We have ten 2-digits packed BCD numbers stored in ROM. Convert it into 20 unpacked BCD numbers and display on a common anode seven segment display connected with port P1. Store these values in ROM: 0x15, 0x 27, 0x 35, 0x 98, 0x 65, 0x 45, 0x 72, 0x 99, 0x 85, 0x 45

Firstly, 1 should be displayed to seven segment, then 5 should be displayed, then 2 then 7 and so on.

Draw circuit diagram of this system.

1 5 2 7
↓ ↓ ↓ ↓
D0 D1 D2 D3 closed switch — 0 (enters)
 opened switch — 1 (leaves)

Question#2: [15 Marks] CLO - 03

We want to implement a car park control system using 8051. Two barriers are used, one at the entry and one at the exit of a car park. When a barrier is lifted to allow a car to pass through, switches are activated which send logic '1' to the microcontroller. When barrier is closed, switch at the barrier gives '0'. When a car enters the parking lot, the system increments a counter, when a car leaves it decrements its counter. LCD is connected to your system that displays the number of spaces available inside the parking lot. Assume maximum 80 cars can be parked inside. If no space is available, then "FULL" should be displayed. Assume that the barriers lift-up automatically when a vehicle approaches them. Also, assume that the entry barrier has a locking mechanism and this mechanism is enabled to lock the barrier so that it does not lift-up when the car park is full. Once barrier is open, only one car can enter at a time.

Draw a diagram of your system showing all the connections with the barriers (switches) and the LCD.

Assume the mechanical switches are connected with the barriers, and they may experience bouncing of 40ms. Incorporate this problem in your software.

Assume that subroutines LCD_Command_Write, LCD_Data_Write, Delay_1msec are already given to you. Accumulator (A) is used to send command to LCD in LCD_Command_Write subroutine. Accumulator (A) is used to write data to LCD in LCD_Data_Write subroutine. Use the available subroutines in your solution. LCD command codes are given on next page.

EL303 – Microprocessor Programming & Interfacing
Quiz#3 (EL-2B/2A)

Roll#: _____
Date: November 22, 2017

Total Marks: 25
Time Allowed: 50mins

Write a program that keeps taking input data from 4x3 keypad until it gets three digits. After getting the input data of three digits, find maximum number in input data array and display it on 2x16 LCD.

Example: If input data entered on keypad is:



2##*7#*3

7

will be displayed on LCD.

Your code must be as generic as possible. Simulate your code in Proteus. Assume that crystal frequency is 11.0592MHz.

Note: It is an open book and open notes quiz. You have to submit your code along with its simulation in Proteus.

0 1 2
3 4 5
6 7 8
* 0 #

8H
9H
0AH

RS1 RS0
1 0
0 0
0 1
1 0 B
1 1

8H, 9H

if 8H > 9H → check 3

else

check 2